

20 Two bags, **A** and **B**, each contain only green marbles and red marbles.

There are 6 green marbles and 4 red marbles in bag **A**.

There are 8 green marbles and 2 red marbles in bag **B**.

One marble is going to be taken at random from bag **A** and placed in bag **B**.

A marble is then going to be taken at random from bag **B**.

Work out the probability that this marble will be a green marble.

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